

Rockchip RV1126B Linux IPC SDK Release Note

ID: RK-FB-YF-A45

Release Version: V1.0.0

Release Date:2025-06-20

Security Level: ☐Top-Secret ☐Secret ☐Internal ☒Public

DISCLAIMER

THIS DOCUMENT IS PROVIDED "AS IS". ROCKCHIP ELECTRONICS CO., LTD.("ROCKCHIP")DOES NOT PROVIDE ANY WARRANTY OF ANY KIND, EXPRESSED, IMPLIED OR OTHERWISE, WITH RESPECT TO THE ACCURACY, RELIABILITY, COMPLETENESS,MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE OR NON-INFRINGEMENT OF ANY REPRESENTATION, INFORMATION AND CONTENT IN THIS DOCUMENT. THIS DOCUMENT IS FOR REFERENCE ONLY. THIS DOCUMENT MAY BE UPDATED OR CHANGED WITHOUT ANY NOTICE AT ANY TIME DUE TO THE UPGRADES OF THE PRODUCT OR ANY OTHER REASONS.

Trademark Statement

"Rockchip", "瑞芯微", "瑞芯" shall be Rockchip's registered trademarks and owned by Rockchip. All the other trademarks or registered trademarks mentioned in this document shall be owned by their respective owners.

All rights reserved. ©2025. Rockchip Electronics Co., Ltd.

Beyond the scope of fair use, neither any entity nor individual shall extract, copy, or distribute this document in any form in whole or in part without the written approval of Rockchip.

Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian,PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

The document presents Rockchip RV1126B Linux IPC SDK, and it is applicable to the RV1126B EVB development board and all IPC/CVR products developed on this board.

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Product Version

Chipset	Kernel Version
RV1126B	Linux 6.1

Revision History

Date	Version	Author	Revision History
2025-03-26	V0.0.1	CWW	Alpha version
2025-05-15	V0.1.0	CWW	Add Specification Differences
2025-06-20	V1.0.0	CWW	Release version

Contents

Rockchip RV1126B Linux IPC SDK Release Note

1. Overview
 - 1.1 Differences between RV1126B and RV1126B-P
 - 1.2 Specification Differences
2. How to Get the SDK
 - 2.1 Get RV1126B Linux IPC SDK
 - 2.1.1 Download Authorization
 - 2.1.2 Get Source Code from Rockchip Code Server
 - 2.1.3 Get Source Code from Local Compression Package
3. Software Development Guide
4. SSH Public Key Operation Introduction
 - 4.1 Key Authority Management

1. Overview

This SDK supports with Linux kernel 6.1 and U-boot v2017.09. It is suitable for RV1126B EVB development boards and all other IPC/CVR products developed based on it.

This SDK provides flexible data path combination interfaces to meet the customized requirements for free combination, please refer to the documents under the project's docs/ directory.

1.1 Differences between RV1126B and RV1126B-P

RV1126B is the full-featured version, which includes functions such as USB3.0, Audio Codec, and Ethernet PHY, etc.

RV1126B-P, on the other hand, is a Pin2Pin version of RV1126. For customers who previously developed products using RV1126, they can directly replace it with RV1126B-P without the need for redesigning the hardware.

1.2 Specification Differences

Specification	RV1126	RV1126B-P (pin to pin)	RV1126B
CPU	4*A7	4*A53	4*A53
Audio	SAI(I2S/PCM/TDM) PDM 2 x DSM(PWM Audio)	SAI(I2S/PCM/TDM) PDM 2 x DSM(PWM Audio)	SAI(I2S/PCM/TDM) PDM 2 x DSM(PWM Audio) 2 x Audio ADC
SARADC	6	6	24
PWM	16	28	28
MIPI CSI	2x4lane	2x4lane	2x4lane/4x2lane
USB	USB2.0 DRD+USB2.0 HOST	USB2.0 DRD+USB2.0 HOST	USB2.0/3.0 DRD+USB2.0 HOST
RTC	N/A	N/A	Built - in
Ethernet	GMAC	GMAC	GMAC+FE PHY

For detailed differences, please refer to the hardware documentation [RV1126&RV1126B-P_Differences_Description_and_Compatibility_Design_Notes_V1.26_20250604_CN&EN.pdf](#).

2. How to Get the SDK

The SDK is released by Rockchip server. Please refer to [Software Development Guide](#) to build a development environment.

2.1 Get RV1126B Linux IPC SDK

2.1.1 Download Authorization

If you have completed the account creating and need to generate your own HTTPS password before downloading the code, please download the repository `https://gerrit.rock-chips.com:8443/repo-release/tools/script` and download the authorization through the `script/Gernerate-Credential.sh` under the repository, this is shown below.

```
mkdir ~/rk_gerrit_password
cd ~/rk_gerrit_password
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/script
cd ./script
./Generate-Credential.x
```

The log of running the authorization script is as follows:

```
Adding configuration for Gerrit server to /home/Test123/rk_gerrit_password
*****
*****
*****
*****
Copyright Statement

Copyright (C) 2024 Rockchip Electronics Co., Ltd. All rights reserved.

BY OPENING OR USING THIS FILE, RECEIVER HEREBY ACKNOWLEDGES AND AGREES THAT THE
SOFTWARE/FIRMWARE AND ITS DOCUMENTATIONS ("ROCKCHIP SOFTWARE") RECEIVED FROM ROCKCHIP ON AN
"AS-IS" BASIS ONLY WITHOUT ANY AND ALL WARRANTIES,
EITHER EXPRESS, IMPLIED OR STATUTORY, INCLUDING, WITHOUT LIMITATION, ANY WARRANTY OR
CONDITION WITH RESPECT TO TITLE, MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE, OR
NON-INFRINGEMENT.

NEITHER DOES ROCKCHIP PROVIDE ANY WARRANTY WHATSOEVER WITH RESPECT TO ANY OPEN SOURCE
TECHNOLOGIES, THIRD-PARTY TECHNOLOGIES OR ANY STANDARD TECHNOLOGIES WHICH MAY BE SUPPORTED
BY, INCORPORATED IN, OR SUPPLIED WITH THE ROCKCHIP SOFTWARE.

RECEIVER EXPRESSLY ACKNOWLEDGES THAT IT IS RECEIVER'S SOLE RESPONSIBILITY TO OBTAIN AND
MAINTAIN ALL NECESSARY LICENSES AND RIGHTS FROM THEIR RESPECTIVE OWNERS TO USE ANY SUCH
THIRD-PARTY TECHNOLOGIES OR ANY STANDARD TECHNOLOGIES.

RECEIVER'S SOLE AND EXCLUSIVE REMEDY AND ROCKCHIP'S ENTIRE AND CUMULATIVE LIABILITY WITH
RESPECT TO THE ROCKCHIP SOFTWARE RELEASED HEREUNDER WILL BE, AT ROCKCHIP 'S OPTION, TO
REVISE OR REPLACE THE ROCKCHIP SOFTWARE AT ISSUE,
OR REFUND ANY FEES OR CHARGE PAID BY RECEIVER TO ROCKCHIP FOR SUCH ROCKCHIP SOFTWARE AT
ISSUE.
```

```

*****
*****
*****
Do you agree to the above terms? (yes/no) yes
Please input your username (make sure to use the username exactly the same as the one in RK
email!!): Test123
Please input your email (make sure to use the company email address!!): Test123@rock-
chips.com
Please input your SSH private key filename (e.g. id_rsa):
Your Credential check has passed
New git credential will be added into your history!!
You are All Set!!

```

2.1.2 Get Source Code from Rockchip Code Server

To get the SDK software package, customers need an account to access the source code repository provided by Rockchip. In order to get code synchronization permission, please provide SSH public key for server authentication and authorization when apply for SDK from Rockchip technical window. About Rockchip server SSH public key authorization, please refer to [SSH Public Key Operation Introduction](#).

Repo, a tool built on Python script by Google to help manage git repositories, is mainly used to download and manage software repository of projects. The download address is as follows:

```

mkdir -p $HOME/repo-tool
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo $HOME/repo-tool

export PATH="$HOME/repo-tool:$PATH"
# test command
repo version

```

The SDK download command is as follows:

```

# Create a SDK dir
mkdir RV1126B_Linux_IPC_SDK
cd RV1126B_Linux_IPC_SDK

# Configure the address of SDK
repo init --repo-url \
    https://gerrit.rock-chips.com:8443/repo-release/tools/repo \
    -u https://gerrit.rock-chips.com:8443/linux/rockchip/ipc/manifests \
    -b rv1126b -m rv1126b_linux_ipc_release.xml

# Download and Sync SDK codes
.repo/repo/repo sync --no-clone-bundle -c --no-tags -j4

```

2.1.3 Get Source Code from Local Compression Package

For quick access to SDK source code, Rockchip Technical Window usually provides corresponding version of SDK initial compression package. In this way, developers can get SDK source code through decompressing the initial compression package, which is the same as the one downloaded by repo.

Take RV1126B_Linux_IPC_SDK_VX.X.X_XXX.tar.bz2 as an example. After getting a initialization package, you can get source code by running the following command:

```
mkdir RV1126B_Linux_IPC_SDK
tar xvf RV1126B_Linux_IPC_SDK_VX.X.X_XXX.tar.bz2 -C RV1126B_Linux_IPC_SDK
cd RV1126B_Linux_IPC_SDK
.repo/repo/repo sync -l
.repo/repo/repo sync --no-clone-bundle -c --no-tags -j4
```

Developers can update via `.repo/repo/repo sync --no-clone-bundle -c --no-tags -j4` command according to update introductions that are regularly released by FAE window.

3. Software Development Guide

For software development, please refer to the quick start documents in the project directory:

```
<SDK>/docs/en/ipc/Rockchip_RV1126B_Quick_Start_Linux_IPC_SDK_EN.pdf
```

4. SSH Public Key Operation Introduction

Please follow the introduction in the “Rockchip_User_Guide_SDK_Application_And_Synchronization_CN” to generate a SSH public key and send the email to sw.fae@rock-chips.com, to get the SDK code.

This document will be released to customers during the process of applying for permission.

4.1 Key Authority Management

Server can monitor download times and IP information of a key in real time. If an abnormality is found, download permission of the corresponding key will be disabled.

Keep the private key file properly. Do not grant second authorization to third parties.